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MID SEMESTER EXAMINATION, SEPTEMBER 2025

Name of the Course: B.Tech Biotechnology

Semester: III

Name of the Paper: Sustainable Agriculture and Organic Farming for Food Security

Paper Code: TBT 306c

Time: 90 minutes

Maximum Marks: 50

Note:-

- (i) All questions are compulsory.
- (ii) Answer any one sub-questions among a & b in each main question.
- (iii) Each question carries 10 marks.

Q1.	(10 Marks)	
(a)	Analyze the interrelationship between agriculture and climate change.	CO1
(b)	Evaluate the role of organic farming in mitigating climate change impacts.	
Q2.	(10 Marks)	
(a)	Define and explain the concept of organic farming along with its guiding principles	CO1
(b)	Assess the major advantages and limitations of organic farming practices.	
Q3.	(10 Marks)	
(a)	Illustrate how sustainable agriculture functions and describe its key components.	CO1
(b)	Examine the challenges of food security and how it is related to UNSDG 2 (Zero Hunger).	
Q4.	(10 Marks)	
(a)	Demonstrate how biotechnology can be applied to address food security concerns.	CO2
(b)	Discuss agroforestry as a system contributing to sustainable agriculture.	
Q5.	(10 Marks)	
(a)	Explain the process, steps, and principles of crop rotation as a farming strategy.	CO2
(b)	Analyze the reasons organic products are often more nutritious and higher priced in markets.	

